Linkmodel: logoffset

Parametrization

This model is an extention of the "log"-link, to include an additional scaled offset, so that

$$y_i \sim \text{Poisson}(\beta x_i + \exp(\eta_i))$$

where $\beta \geq 0$ is an optional hyperparameter (default fixed to be 1), x_i is a (required) non-negative covariate.

Hyperparameters

The parameter β represented as

$$\theta = \log \beta$$

and the prior is defined on θ.

Specification

Use `model="logoffset"` within `control.link`. The `variant=0` in `control.link` gives the link function above, while `variant=1` gives $\beta x_i - \exp(\eta_i)$ (for other likelihoods).

Hyperparameter spesification and default values

doc `Log-link with an offset`

hyper

theta

hyperid `49001`
name `beta`
short.name `b`
prior `normal`
param `0 100`
initial `0`
fixed `TRUE`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

pdf `logoffset`

Example

```
n = 1000
off = rpois(n, lambda = 1)
x = rnorm(n)
lambda = off + exp(1 + x)
y = rpois(n, lambda = lambda)

r = inla(y ~ 1 + x, data = data.frame(y, x, link.cov = off),
        family = "poisson",
        control.family = list(
                control.link = list(
```

```
                        model = "logoffset",
                        hyper = list(
                                beta = list(
                                        initial = 0,
                                        ##fixed = TRUE
                                        fixed = FALSE
                                        ))),
        link.covariates = link.cov)
```

Notes